

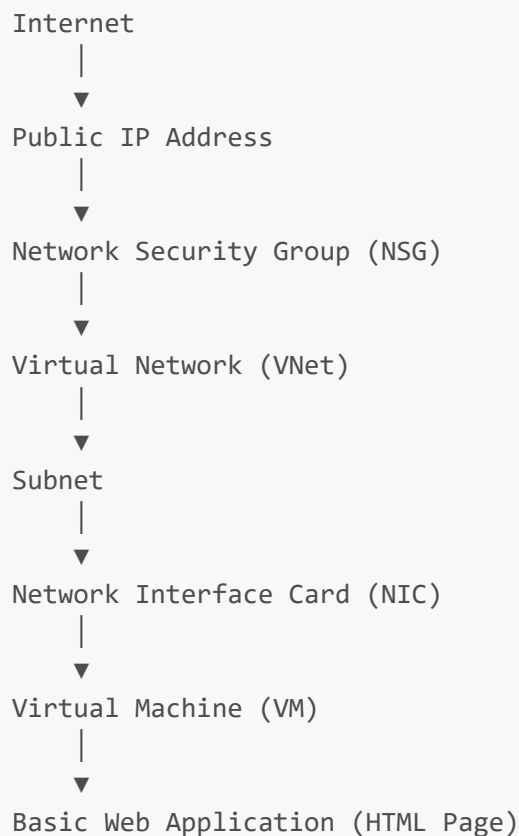
Azure Virtual Machine Deployment - Manual Implementation Guide

Complete Step-by-Step Process via Azure Portal

Prerequisites

- Active Azure Subscription
 - Azure Portal Access: portal.azure.com
 - Basic understanding of networking concepts
-

Architecture Overview



Phase 1: Create Resource Group

Step 1: Navigate to Resource Groups

Azure Portal Homepage
→ Click "Resource groups" (left sidebar or search bar)

→ Click "+ Create" button

Step 2: Configure Resource Group

Project Details:

Subscription	: [Your Subscription]
Resource Group	: my-vm-resource-group
Region	: East US

Step 3: Review and Create

- Click "Review + Create"
- Click "Create"
- Wait for deployment confirmation ✓

Phase 2: Create Virtual Network (VNet)

Step 1: Navigate to Virtual Networks

Azure Portal

- Search "Virtual Networks" in search bar
- Click "Virtual Networks"
- Click "+ Create"

Step 2: Configure Basics Tab

BASICS TAB

Subscription	: [Your Subscription]
Resource Group	: my-vm-resource-group
Name	: my-virtual-network
Region	: East US

Step 3: Configure IP Addresses Tab

IP ADDRESSES TAB

```
IPv4 address space : 10.0.0.0/16
```

```
Subnet Configuration:
```

```
→ Click "+ Add subnet"
```

```
Subnet Name      : my-subnet
```

```
Subnet Range     : 10.0.1.0/24
```

```
→ Click "Add"
```

Step 4: Security Tab (Skip/Default)

```
→ Leave DDoS Protection: Basic
```

```
→ Leave Firewall: Disabled
```

```
→ Click "Next: Tags"
```

Step 5: Create VNet

```
→ Click "Review + Create"
```

```
→ Verify all settings
```

```
→ Click "Create"
```

```
→ Wait for deployment ✓
```

Phase 3: Create Public IP Address

Step 1: Navigate to Public IP Addresses

```
Azure Portal
```

```
→ Search "Public IP addresses" in search bar
```

```
→ Click "Public IP addresses"
```

```
→ Click "+ Create"
```

Step 2: Configure Public IP

```
PUBLIC IP CONFIGURATION
```

```
Subscription      : [Your Subscription]
```

```
Resource Group    : my-vm-resource-group
```

```
Region            : East US
```

```
Name              : my-public-ip
```

```
IP Version        : IPv4
```

```
SKU                : Standard
```

Tier	: Regional
Assignment	: Static
Routing Pref	: Microsoft Network
Idle Timeout	: 4 minutes
DNS Name Label	: my-vm-app-demo (optional)

Step 3: Create Public IP

- Click "Review + Create"
- Click "Create"
- Wait for deployment ✓
- Note the assigned Public IP Address

Phase 4: Create Network Security Group (NSG)

Step 1: Navigate to NSG

- Azure Portal
- Search "Network Security Groups"
 - Click "Network Security Groups"
 - Click "+ Create"

Step 2: Configure NSG Basics

NSG BASICS

Subscription	: [Your Subscription]
Resource Group	: my-vm-resource-group
Name	: my-nsg
Region	: East US

- Click "Review + Create"
- Click "Create"

Step 3: Configure Inbound Security Rules

- After NSG is created:
- Go to "my-nsg" resource

- Click "Inbound security rules" (left sidebar)
- Click "+ Add"

Rule 1: Allow SSH (Port 22)

INBOUND RULE - SSH

Source	: Any
Source Port	: *
Destination	: Any
Service	: SSH
Destination Port	: 22
Protocol	: TCP
Action	: Allow
Priority	: 100
Name	: Allow-SSH

- Click "Add"

Rule 2: Allow HTTP (Port 80)

INBOUND RULE - HTTP

Source	: Any
Source Port	: *
Destination	: Any
Service	: HTTP
Destination Port	: 80
Protocol	: TCP
Action	: Allow
Priority	: 110
Name	: Allow-HTTP

- Click "Add"

Rule 3: Allow HTTPS (Port 443)

INBOUND RULE - HTTPS

Source	: Any
Source Port	: *
Destination	: Any
Service	: HTTPS

Destination Port:	443
Protocol	: TCP
Action	: Allow
Priority	: 120
Name	: Allow-HTTPS

→ Click "Add"

Phase 5: Create Virtual Machine

Step 1: Navigate to Virtual Machines

Azure Portal

- Click "Virtual Machines" (left sidebar)
- Click "+ Create"
- Select "Azure Virtual Machine"

Step 2: Configure BASICS Tab

PROJECT DETAILS

Subscription	: [Your Subscription]
Resource Group	: my-vm-resource-group

INSTANCE DETAILS

VM Name	: my-linux-vm
Region	: East US
Availability	: No infrastructure redundancy
Security Type	: Standard
Image	: Ubuntu Server 22.04 LTS
VM Architecture	: x64
Size	: Standard_B1s (1vCPU, 1GB)
	→ Click "See all sizes"
	→ Search "B1s"
	→ Select Standard_B1s

ADMINISTRATOR ACCOUNT

Auth Type	: Password
Username	: azureuser
Password	: MySecurePass@123
Confirm Password	: MySecurePass@123

INBOUND PORT RULES

Public Inbound : Allow selected ports
Select Ports : HTTP(80), HTTPS(443), SSH

Step 3: Configure DISKS Tab

DISK OPTIONS

OS Disk Type : Standard SSD
OS Disk Size : 30 GiB (Default)
Encryption : Default

→ Click "Next: Networking"

Step 4: Configure NETWORKING Tab

NETWORK INTERFACE

Virtual Network : my-virtual-network
Subnet : my-subnet (10.0.1.0/24)
Public IP : my-public-ip
NIC NSG : Advanced
Configure NSG : my-nsg
Delete NIC with : ✓ Checked
Delete IP with : ✓ Checked

LOAD BALANCING

Load Balancing : None

Step 5: Configure MANAGEMENT Tab

MONITORING

Boot Diagnostics: Enable with managed account

AUTO-SHUTDOWN

Enable Auto-shutdown: Enabled (Cost Saving)

Shutdown Time : 11:00 PM

Time Zone : Your Time Zone

Step 6: Configure ADVANCED Tab

- Leave all settings as Default
- Click "Next: Tags"

Step 7: Add Tags (Optional)

TAGS

Name : Environment Value: Development

Name : Project Value: VM-Demo

Name : Owner Value: YourName

Step 8: Review + Create

- Click "Review + Create"
- Review all configuration summary
- Click "Create"
- Wait for deployment (3-5 minutes) ✓

Phase 6: Verify VM Deployment

Step 1: Check VM Status

Azure Portal

- Go to "Virtual Machines"
- Click "my-linux-vm"
- Verify Status: Running ✓
- Note the Public IP Address

Step 2: Verify Network Configuration

In VM Overview Page:

VM DETAILS TO VERIFY	
Status	: Running
Public IP	: XX.XX.XX.XX (your IP)
Private IP	: 10.0.1.4 (auto-assigned)
Virtual Network	: my-virtual-network
Subnet	: my-subnet

Phase 7: Connect to VM and Host Application

Step 1: Connect via Azure Portal (Browser SSH)

Azure Portal

- Go to "my-linux-vm"
- Click "Connect" (top menu)
- Select "SSH"
- Click "Open Azure Cloud Shell"

OR

- Select "Connect via Bastion" (if configured)

OR use any SSH Client:

- Copy the SSH command shown
- Open any SSH client (PuTTY, etc.)

Step 2: Using Azure Serial Console (Alternative)

Azure Portal

- Go to "my-linux-vm"
- Click "Serial Console" (left sidebar under Support)
- Wait for console to load
- Login with credentials

Step 3: Install Web Server via Azure Run Command

Azure Portal

- Go to "my-linux-vm"

- Click "Run command" (left sidebar)
- Click "RunShellScript"

Script to Run in Run Command:

```
#!/bin/bash

# Update package list
apt-get update -y

# Install Nginx Web Server
apt-get install nginx -y

# Start and enable Nginx
systemctl start nginx
systemctl enable nginx

# Create custom HTML page
cat > /var/www/html/index.html << 'EOF'
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Azure VM - Web Application</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }
    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background: linear-gradient(135deg, #0078d4 0%, #00bcf2 100%);
      min-height: 100vh;
      display: flex;
      align-items: center;
      justify-content: center;
    }
    .container {
      background: white;
      border-radius: 20px;
      padding: 50px;
      text-align: center;
      box-shadow: 0 20px 60px rgba(0,0,0,0.3);
      max-width: 700px;
      width: 90%;
    }
    .azure-logo {
      font-size: 80px;
      margin-bottom: 20px;
    }
```

```
}
h1 {
  color: #0078d4;
  font-size: 2.5em;
  margin-bottom: 15px;
}
.status-badge {
  background: #00b050;
  color: white;
  padding: 8px 20px;
  border-radius: 20px;
  font-size: 0.9em;
  display: inline-block;
  margin-bottom: 30px;
}
.info-grid {
  display: grid;
  grid-template-columns: 1fr 1fr;
  gap: 15px;
  margin: 30px 0;
}
.info-card {
  background: #f3f9ff;
  border: 2px solid #0078d4;
  border-radius: 10px;
  padding: 20px;
}
.info-card h3 {
  color: #0078d4;
  margin-bottom: 8px;
  font-size: 0.85em;
  text-transform: uppercase;
}
.info-card p {
  color: #333;
  font-size: 1em;
  font-weight: bold;
}
.footer {
  margin-top: 30px;
  color: #888;
  font-size: 0.85em;
}
</style>
</head>
<body>
  <div class="container">
    <div class="azure-logo">  </div>
    <h1>Azure VM Successfully Deployed!</h1>
    <div class="status-badge">✓ Application Running</div>
    <p style="color:#555; margin-bottom:20px;">
      This web application is hosted on Microsoft Azure Virtual Machine
    </p>
    <div class="info-grid">
```

```
<div class="info-card">
  <h3>Cloud Provider</h3>
  <p>Microsoft Azure</p>
</div>
<div class="info-card">
  <h3>Web Server</h3>
  <p>Nginx</p>
</div>
<div class="info-card">
  <h3>Operating System</h3>
  <p>Ubuntu 22.04 LTS</p>
</div>
<div class="info-card">
  <h3>Status</h3>
  <p style="color:#00b050">● Live & Running</p>
</div>
</div>
<div class="footer">
  <p>Deployed via Azure Portal | Azure Virtual Machine Demo</p>
</div>
</div>
</body>
</html>
EOF

# Set correct permissions
chmod 644 /var/www/html/index.html

# Restart Nginx to apply changes
systemctl restart nginx

echo "Web application deployed successfully!"
```

- Paste script in the script box
- Click "Run"
- Wait for execution (2-3 minutes)
- Verify output: "Web application deployed successfully!"

Phase 8: Verify Public Access

Step 1: Get Public IP

- Azure Portal
- Go to "my-linux-vm"
 - Overview page
 - Copy "Public IP Address": XX.XX.XX.XX

Step 2: Test Application

Open any Web Browser:

- Type: `http://XX.XX.XX.XX`
- Press Enter
- You should see the Azure VM Web Application page ✓

Step 3: Verify in Azure Portal

Azure Portal - Verify Connectivity:

- VM → Networking → Check NSG rules active
- VM → Overview → Status: Running
- Public IP → Associated to NIC ✓

Phase 9: Monitor Your VM

Step 1: Check VM Metrics

Azure Portal

- Go to "my-linux-vm"
- Click "Metrics" (left sidebar)
- Add Metric:
 - Percentage CPU
 - Network In Total
 - Network Out Total
 - Disk Read Bytes

Step 2: Check Activity Log

Azure Portal

- Go to "my-linux-vm"
- Click "Activity log" (left sidebar)
- View all operations performed

Step 3: Check Boot Diagnostics

Azure Portal

- Go to "my-linux-vm"
- Click "Boot diagnostics" (left sidebar)
- Click "Screenshot" to verify VM boot

Resource Summary

DEPLOYED RESOURCES SUMMARY	
Resource Type	Name
Resource Group	my-vm-resource-group
Virtual Network	my-virtual-network (10.0.0.0/16)
Subnet	my-subnet (10.0.1.0/24)
Public IP	my-public-ip (Static, Standard)
NSG	my-nsg (SSH, HTTP, HTTPS rules)
Virtual Machine	my-linux-vm (Ubuntu 22.04, Standard_B1s)
OS Disk	my-linux-vm_OsDisk (Standard SSD)
NIC	my-linux-vm-nic
Web Server	Nginx
Application	Custom HTML Web Page

DHCP & IP Assignment Explanation

HOW IP ASSIGNMENT WORKS IN AZURE
PRIVATE IP (Internal): • Azure DHCP automatically assigns private IP • Range: 10.0.1.0/24 → VM gets 10.0.1.4 • First 3 IPs (.1, .2, .3) reserved by Azure • VM does NOT know its public IP PUBLIC IP (External): • Assigned as Static (won't change on restart) • Mapped to NIC via NAT by Azure • Traffic flows: Internet → Public IP → NAT → Private IP Flow: User Browser → XX.XX.XX.XX:80 (Public IP) → Azure NAT → 10.0.1.4:80 (Private IP) → VM Nginx → Serves HTML Page → Response back to User ✓

Cost Management

Stop VM When Not in Use

Azure Portal

- Go to "my-linux-vm"
- Click "Stop" (top menu)
- Confirm stop
- Status changes to: Stopped (Deallocated)
- Compute charges STOP ✓
- Storage charges continue (minimal)

Delete All Resources (Cleanup)

Azure Portal

- Go to "my-vm-resource-group"
- Click "Delete resource group"
- Type resource group name to confirm
- Click "Delete"
- ALL resources deleted ✓ (No more charges)

Troubleshooting

ISSUE	SOLUTION
Can't access HTTP	Check NSG - Port 80 rule exists Check VM is Running state Verify Public IP is associated
SSH not working	Check NSG - Port 22 rule exists Verify correct username/password
Website not loading	Run command - check nginx status Verify nginx is running
VM not starting	Check Boot Diagnostics Check Activity Log for errors

✓ Deployment Complete!

SUCCESS CHECKLIST:

- ☐ Resource Group Created
- ☐ Virtual Network Created with Subnet
- ☐ Public IP Address Created (Static)
- ☐ NSG Created with SSH, HTTP, HTTPS Rules

- ❑ Virtual Machine Deployed (Ubuntu 22.04)
- ❑ Nginx Web Server Installed
- ❑ Custom HTML Application Hosted
- ❑ Application Accessible via Public IP
- ❑ `http://XX.XX.XX.XX` → Shows Web Page ✓